

# SHAD ALI SHAH

Economics & Economic Research | Econometrics | Data Science | Data Analytics | Business Intelligence  
Islamabad, Pakistan | +92-334-890-1607 | shadalishah45@gmail.com | LinkedIn | GitHub | Portfolio

## Professional Summary

Entry-level Economics and Data professional with **MPhil Economics coursework and thesis completed**, a **Data Analytics specialization**, Graduate Research Assistant experience, three analytics internships, and 29 multidisciplinary projects. Proficient in **Python, SQL, R, Stata, EViews, Power BI, and Tableau**, with experience in econometrics, data analysis, statistical modelling, machine learning, forecasting, and analytical reporting.

## Core Competencies

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|--------------------------------|---|---------------------------------------------------------------------------------------------------------------------------------------|
| <b>Economics Research</b>      | & | Macroeconomics, microeconomics, development economics, financial economics, policy analysis, literature review, quantitative research |
| <b>Econometrics Statistics</b> | & | VAR, SVAR, VECM, ARIMA, ADF/KPSS tests, Johansen cointegration, Granger causality, IRFs, FEVD, statistical modelling                  |
| <b>Data Science &amp; ML</b>   |   | Python, Pandas, NumPy, Scikit-learn, regression, classification, Random Forest, K-Means, feature engineering, model evaluation        |
| <b>Analytics &amp; BI</b>      |   | SQL, Google BigQuery, Excel, Power BI, Tableau, data cleaning, EDA, forecasting, KPI reporting, dashboard development                 |
| <b>Tools</b>                   |   | R, Stata, EViews, Jupyter Notebook, Google Colab, Git/GitHub, Quarto, LaTeX                                                           |

## Research & Professional Experience

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| <b>Graduate Research Assistant</b><br><i>Quaid-i-Azam University</i> | February 2026 - Present<br>Islamabad |
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- Review economic literature and organize evidence to support faculty-supervised research and structured documentation.
- Collect, clean, and maintain research datasets to support quantitative analysis and reproducible analytical workflows.
- Analyse research data and prepare tables, visualizations, findings summaries, and structured analytical outputs.
- Document research methods and contribute to review activities under faculty supervision.

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| <b>Data Science Intern</b><br><i>Arch Technologies</i> | 1 May 2026 - 30 June 2026<br>Remote |
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- Analysed and preprocessed datasets with Python and Pandas to produce modelling-ready data for predictive workflows.
- Applied K-Means clustering to segment customers and produce interpretable behavioural groups.
- Built and evaluated a Random Forest workflow with Scikit-learn and produced documented model findings.
- Presented analytical methods and results through structured reports for stakeholder review.

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| <b>Data Science Intern</b><br><i>CodeAlpha</i> | 10 May 2026 - 10 June 2026<br>Virtual |
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- Developed advertising-sales and used-car-price regression workflows with Python, Pandas, and Scikit-learn.
- Evaluated baseline and ensemble models with MAE, RMSE, and R-squared to produce comparative performance findings.
- Automated CSV integration, cleaning, deduplication, and standardization to produce consistent, reusable datasets.
- Documented EDA, modelling, evaluation, and forecasting outputs to support reproducible analysis.

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| <b>AI/ML Engineering Intern - Best Intern Award</b><br><i>DevelopersHub Corporation</i> | 11 May 2026 - 22 June 2026<br>Remote |
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- Built Python classification and regression pipelines covering preprocessing, feature engineering, and model testing.
- Evaluated models with appropriate performance metrics and produced documented methods and results.
- Translated technical findings into clear, decision-ready explanations through stakeholder reports.
- Earned the Best Intern Award for analytical performance, documentation quality, and timely delivery.

## Independent Economics, Data & Analytics Experience | GitHub | 2025-2026

- Conducted self-directed research across macroeconomic, employment, financial, customer, and transaction datasets to produce multidisciplinary analyses.
- Applied econometric, statistical, and machine-learning methods to produce models, forecasts, classifications, and customer segments.
- Used Python, R, SQL, Stata, EViews, BigQuery, Power BI, and Tableau to produce cleaned datasets, models, KPI analyses, and visualizations.
- Developed reproducible code, notebooks, dashboards, forecasts, analytical tables, research summaries, and documented GitHub repositories. [github.com/shadalishah](https://github.com/shadalishah).

## Selected Economics, Data Science & Analytics Projects

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### Pakistan Macroeconomic Dynamics

GitHub

*VAR, SVAR, VECM, ARIMA, R, Stata, Quarto*

- Analysed 44+ quarters of GDP, government spending, tax revenue, CPI inflation, and policy-rate data.
- Applied stationarity, cointegration, causality, impulse-response, and variance-decomposition methods.
- Produced a reproducible workflow supporting interpretation of short-run and long-run macroeconomic relationships.

### Pakistan E-Commerce Analytics

GitHub

*Python, Pandas, RFM, K-Means, forecasting*

- Cleaned more than one million raw records into **584,524 transactions** covering 115,326 customers.
- Applied RFM segmentation and K-Means clustering to identify customer groups and retention insights.
- Developed sales forecasts and documented transaction, customer, and market patterns in a reproducible workflow.

### Advanced SQL Analytics with Google BigQuery

GitHub

*SQL, CTEs, window functions, customer analytics*

- Queried and segmented **50,000 customer records** using SQL, CTEs, and window functions in BigQuery.
- Conducted data-quality, churn, and customer lifetime-value analysis to identify business patterns.
- Delivered executive KPI summaries and an action matrix to communicate analytical priorities.

### Pakistan and Global Employment Analytics

GitHub

*World Bank data, Tableau, forecasting, business intelligence*

- Analysed **57,500+ observations across 183 countries** covering employment trends from 1991-2025.
- Developed forecasts to 2030 and examined gender and cross-country labour-market patterns.
- Built an interactive Tableau dashboard highlighting policy-oriented trends and decision-ready findings.

## 29 Economics, Data Science & Analytics Projects | 2025-2026

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Completed a portfolio of **29 independent, academic/research, and internship projects** spanning macroeconomics, microeconomics, econometrics, financial economics, statistical learning, machine learning, Python, SQL, R, EDA, feature engineering, regression, classification, clustering, forecasting, business intelligence, and data visualization. Repositories include code, notebooks, analytical outputs, dashboards, visualizations, and documentation where available. Full portfolio: [github.com/shadalishah](https://github.com/shadalishah).

## Education

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### MPhil Economics - Coursework and Thesis Completed

2025 - 2026

*Quaid-i-Azam University, Islamabad*

**CGPA: 3.50/4.00**

Completed all coursework and finished the thesis before the applicable deadline under faculty supervision. The thesis has written supervisor approval; formal university submission is pending until the university's prescribed submission period. Focus: econometrics, macroeconomics, quantitative research, time-series analysis, and data science. **Available for full-time employment.**

### BS Economics - Graduated with Distinction

2020 - 2024

*Kohat University of Science and Technology*

**CGPA: 3.89/4.00**

**Relevant Coursework:** Econometrics, Statistics, Mathematical Economics, Macroeconomics, Microeconomics, Development Economics, Monetary Economics, Financial Institutions and Markets, Public Finance and Fiscal Policy, Research Methodology

**Undergraduate Thesis:** The Influence of Biofuel Consumption on Environmental Sustainability in Developed Countries

## Certifications

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- IBM Data Science Professional Certificate – IBM;
- Machine Learning – DeepLearning.AI.
- Google Data Analytics; Google Advanced Data Analytics; Google Business Intelligence – Google.
- Generative AI Data Analyst – Vanderbilt University;
- Microsoft AI & ML Engineering – Microsoft.
- Survey Data Collection and Analytics;– University of Michigan
- Python and Statistics for Financial Analysis;–(HKUST)
- Regression and Modelling with Stata.– Pack